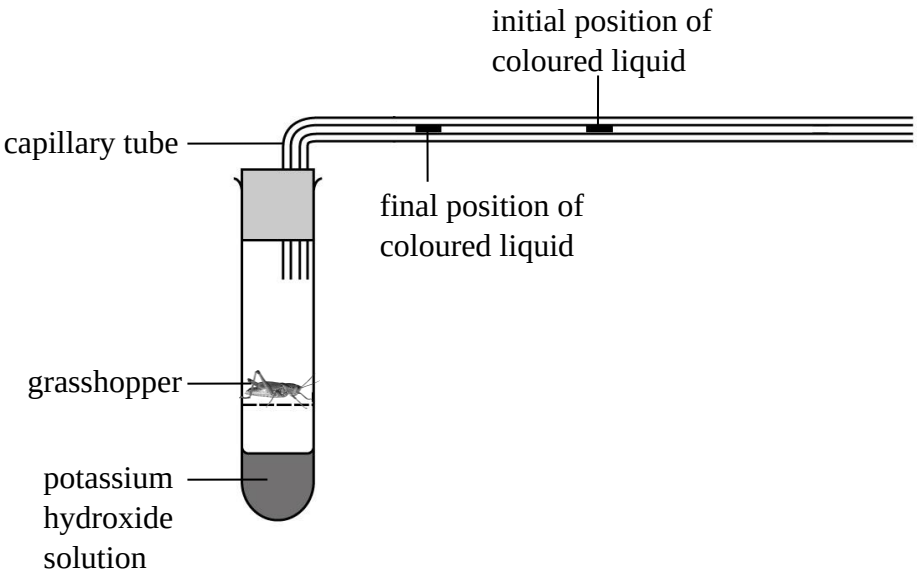
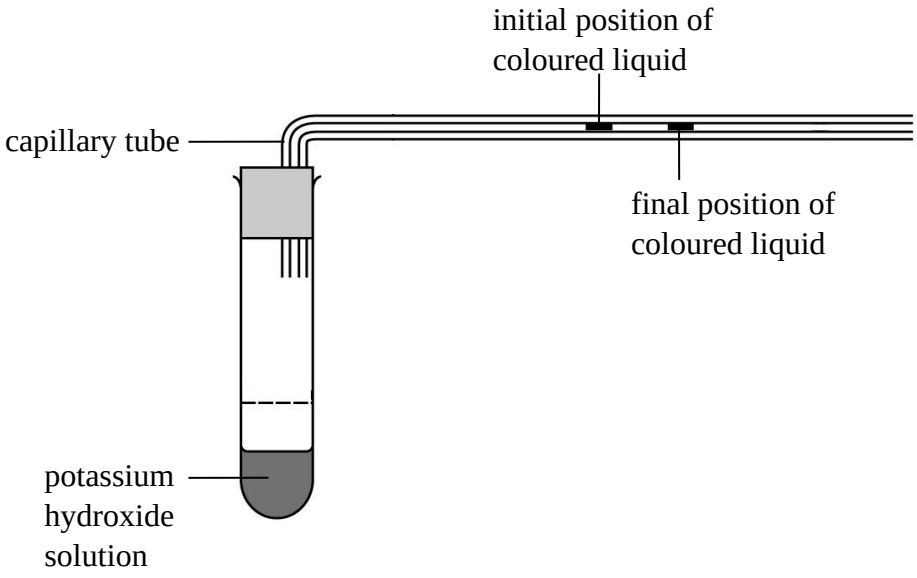
Multiple-choice questions

|!|BCEMB0220000001|!| ★ SI

Directions: The following three questions refer to the experiment below, which is conducted to investigate the respiration of a grasshopper.



Set-up A



Set-up B

Which of the following correctly states the function of potassium hydroxide solution in the experiment?

A. to release carbon dioxide
B. to absorb carbon dioxide
C. to release oxygen
D. to absorb water vapour

□

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##B##

Skill assessed	Mark
Practical skills	1

|!|BCEMB0220000002|!| ★ **SI**

The movement of the coloured liquid in set-up A was due to

A. release of water vapour from the grasshopper.
B. release of carbon dioxide from the grasshopper.
C. uptake of oxygen by the grasshopper.
D. absorption of oxygen by the potassium hydroxide solution.

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0220000003|!| ★ **SI**

Which of the following correctly explains the results of set-up B?

A. release of carbon dioxide from the potassium hydroxide solution
B. absorption of oxygen by the potassium hydroxide solution
C. a decrease in environmental temperature
D. an increase in environmental temperature

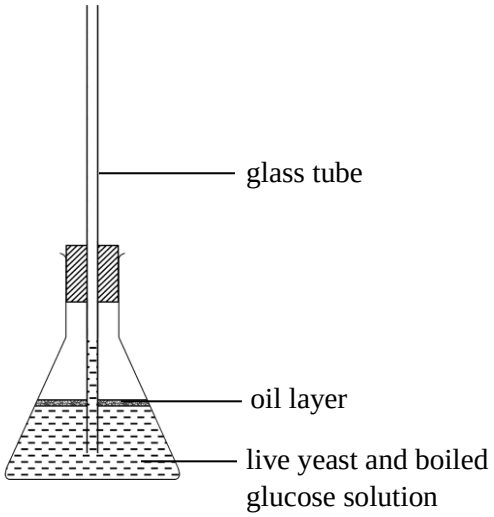
##D##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMB0220000004|!| ★ **SI**

Directions: The following two questions refer to the set-up below, which is used to study the action of yeast on glucose.



Which of the following correctly states and explains the change in the liquid level in the glass tube after 30 minutes?

	Change in liquid level	***Cause for the change***
A.	rises	Alcohol is released.
B.	rises	Carbon dioxide is released.
C.	drops	Oxygen is consumed.
D.	drops	Glucose is consumed.

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0220000005|!| ★ **SI**

A control set-up is needed in order to show that living yeast is essential for causing a change in the liquid level in the glass tube. The control-set-up should contain

A. boiled glucose solution only.
B. living yeast and boiled water.
C. dead yeast and boiled water.
D. dead yeast and boiled glucose solution.

##D##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMB0220000006|!| ★ **Cross topic**

Which of the following correctly compares photosynthesis and aerobic respiration?

	Photosynthesis	***Aerobic respiration***
A.	catabolic	anabolic
B.	no ATP is produced	ATP is produced
C.	electron transport is not involved	electron transport is involved
D.	uses water as a reactant in the first stage	forms water as a product in the final stage

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0220000007|!|

Which of the following processes produce(s) carbon dioxide as an end product?

(1) aerobic respiration
(2) lactic acid fermentation
(3) alcoholic fermentation

A. (1) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0220000008|!| **Cross topic**

Which of the following processes are supported by the energy released from the hydrolysis of ATP?

(1) synthesis of enzymes
(2) cell division
(3) water absorption in the alimentary canal

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0220000009|!| ★

Respiration ***cannot*** take place in

(1) the xylem vessel.
(2) the sperm.
(3) the enamel of teeth.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (1) and (3) only

##D##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0220000010|!| **Cross topic**

Which of the following processes are inhibited when no ATP is available?

(1) photolysis of water
(2) contraction of skeletal muscles
(3) transmission of nerve impulses via nerves

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0220000011|!| **Cross topic**

Which of the following processes in plants does ***not*** require energy from ATP?

A. transpiration
B. mitosis
C. absorption of nitrates from soil
D. translocation of sucrose

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0220000012|!| ★

Which of the following statements about ATP is/are correct?

(1) ATP can be transported from cell to cell.
(2) ATP is synthesized by adding a phosphate group to ADP.
(3) ATP is produced in mitochondria only.

A. (1) only
B. (2) only
C. (1) and (2) only
D. (1) and (3) only

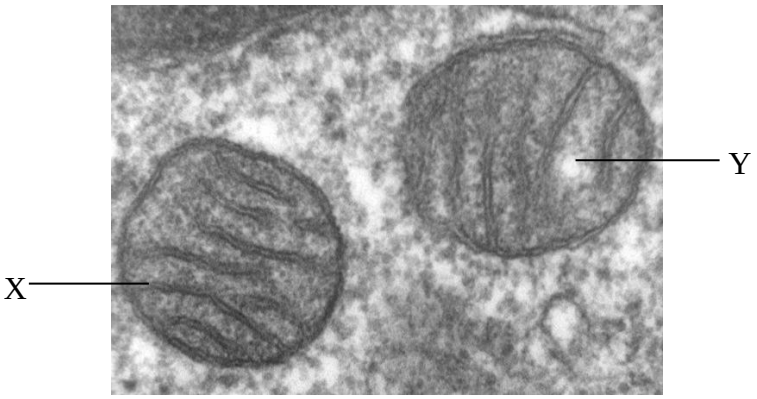
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##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMA0220000013|!| ★★

The following electron micrograph shows two mitochondria.



(Image source: https://commons.wikimedia.org/wiki/File:Mitochondria,_mammalian_lung_-_TEM.jpg)

Which of the following correctly shows the stages of respiration which occur in sites X and Y respectively?

	Site X	***Site Y***
A.	Krebs cycle	glycolysis
B.	Krebs cycle	oxidative phosphorylation
C.	oxidative phosphorylation	glycolysis
D.	oxidative phosphorylation	Krebs cycle

□

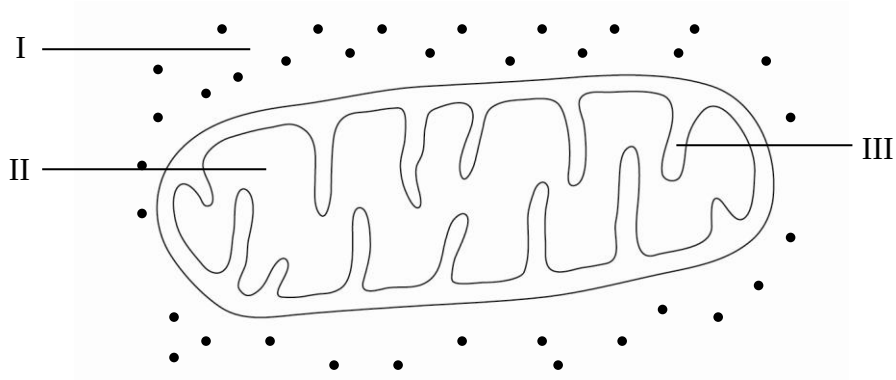
##D##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0220000014|!| ★

The following diagram shows a mitochondrion and the surrounding cytoplasm.



In which of the following sites is carbon dioxide produced?

A. site I only
B. site II only
C. sites I and II only
D. sites II and III only

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0220000015|!|

Which of the following are the products of glycolysis?

(1) acetyl-CoA
(2) ATP
(3) NADH

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0220000016|!|

How many molecules of ATP, NADH and pyruvate are produced from a glucose molecule in glycolysis?

	ATP	*NADH*	*Pyruvate*
A.	2	3	2
B.	2	2	2
C.	2	1	1
D.	4	2	1

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0220000017|!|

Which of the following reactions of respiration consume(s) ATP?

(1) glucose ⟶ triose phosphate
(2) triose phosphate ⟶ pyruvate
(3) pyruvate ⟶ acetyl-CoA

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

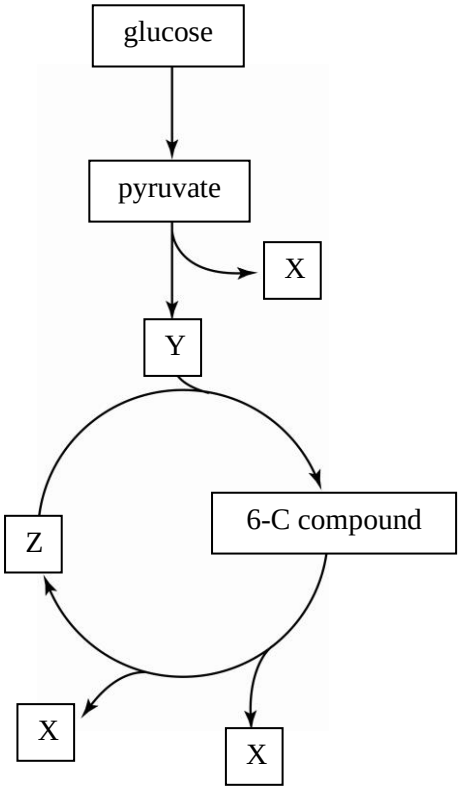
##A##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0220000018|!| ★

The diagram below shows an outline of respiration. X, Y and Z are different types of carbon-containing substances.



Which of the following combinations correctly shows the number of carbon atom(s) in X, Y and Z?

	X	***Y***	***Z***
A.	1	2	4
B.	1	3	3
C.	2	3	4
D.	2	1	3

□

##A##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0220000019|!| ★

At which of the following stages of respiration is carbon dioxide released?

(1) glycolysis
(2) conversion of pyruvate to acetyl-CoA
(3) Krebs cycle

A. (3) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0220000020|!|

Where does the Krebs cycle occur in a cell?

(1) cytoplasm
(2) inner membrane of mitochondrion
(3) matrix of mitochondrion

A. (1) only
B. (2) only
C. (3) only
D. (2) and (3) only

##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0220000021|!|

Which of the following reactions take(s) place in the Krebs cycle?

(1) conversion of pyruvate to acetyl-CoA
(2) combination of acetyl-CoA with a 4-C compound
(3) reduction of NAD to NADH

A. (2) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0220000022|!|

Which of the following processes is ***not*** involved in oxidative phosphorylation?

A. regeneration of NAD
B. formation of water
C. formation of carbon dioxide
D. formation of ATP

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0220000023|!| ★

What is the role of NADH in aerobic respiration?

A. to combine with oxygen to form water
B. to reduce intermediates in the Krebs cycle
C. to accept electrons from a series of electron carriers in the inner membrane of the mitochondrion
D. to transfer hydrogen to a series of electron carriers in the inner membrane of the mitochondrion

##D##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0220000024|!|

Which of the following reactions take(s) place in oxidative phosphorylation?

(1) formation of water
(2) oxidation of FADH to FAD
(3) hydrolysis of ATP

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0220000025|!| ★ **Cross topic**

Which of the following processes produce(s) ATP?

(1) Calvin cycle
(2) glycolysis
(3) oxidative phosphorylation

A. (3) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

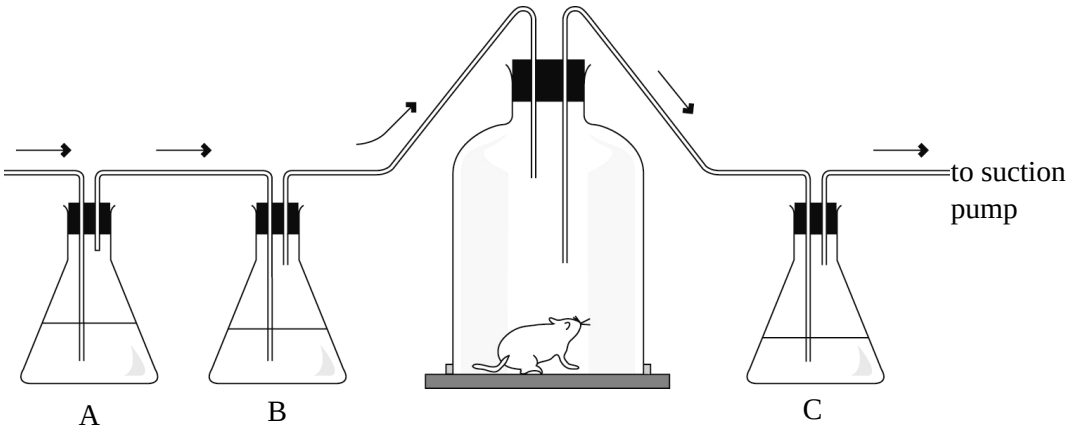
##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0220000026|!| **SI**

Directions: The following two questions refer to the diagram below which shows a set-up used to find out whether carbon dioxide is released by a living mouse.



What does flask A contain?

A. potassium hydroxide solution
B. hydrochloric acid
C. lime water
D. hydrogencarbonate indicator solution

##A##

Skill assessed	Mark
Practical skills	1

|!|BCEMC0220000027|!| **SI**

Flasks B and C contain lime water. Which of the following shows the appearance of the lime water in flasks B and C after 15 minutes?

	Flask B	***Flask C***
A.	milky	milky
B.	milky	colourless
C.	colourless	milky
D.	colourless	colourless

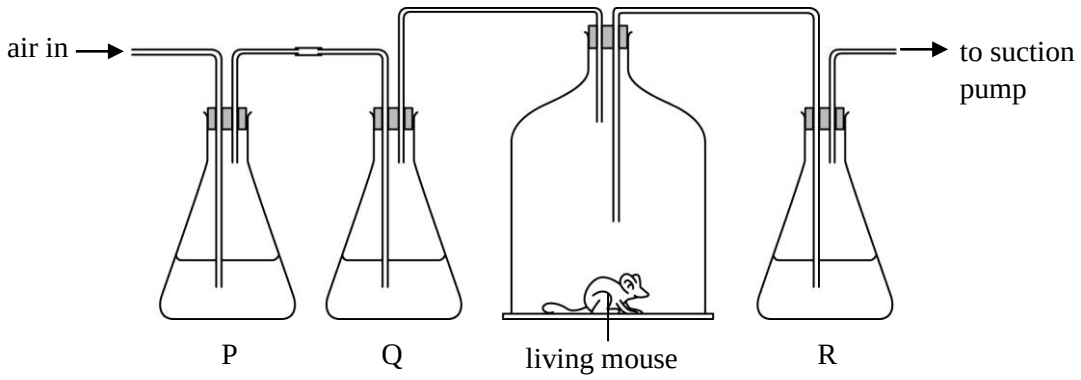
##C##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMB0220000028|!| ★ SI

Directions: The following three questions refer to the diagram below which shows a set-up used to investigate respiration in a living mouse.



Which of the following correctly shows the appearance of the solutions in the flasks at the end of the experiment?

	P	***Q***	***R***
A.	colourless	colourless	milky
B.	colourless	colourless	colourless
C.	milky	colourless	milky
D.	colourless	milky	milky

##A##

Skill assessed	Mark
Practical skills	1

|!|BCEMB0220000029|!| ★ SI

Which of the following is a possible hypothesis of this investigation?

A. A living mouse oxidizes food substances.
B. A living mouse releases carbon dioxide.
C. A living mouse takes in oxygen.
D. A living mouse releases heat.

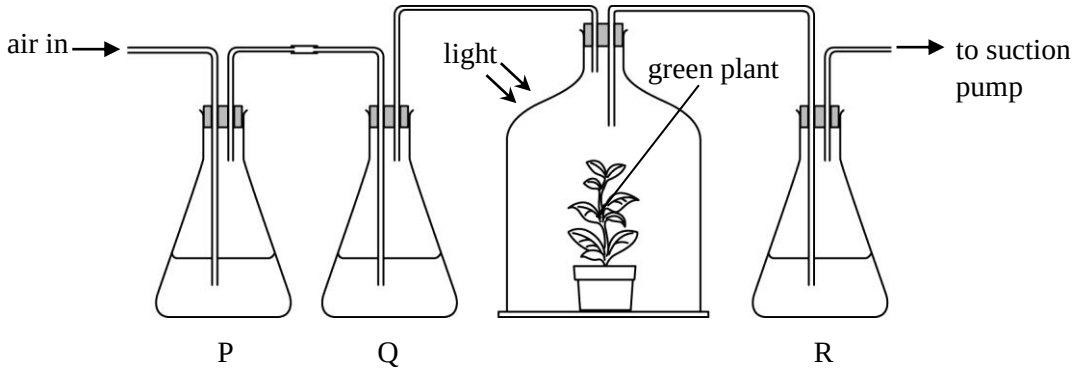
##B##

Skill assessed	Mark
Practical skills	1

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|!|BCEMA0220000030|!| ★★ **SI**

A student used the same set-up to demonstrate the release of carbon dioxide by a potted green plant, as shown in the diagram below.



The above set-up was not set up properly. What corrections should be made to the set-up in order to fulfill the aim of the investigation?

(1) cover the bell jar with a piece of black cloth
(2) wrap the pot in a plastic bag
(3) wrap the bell jar with cotton wool

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

□

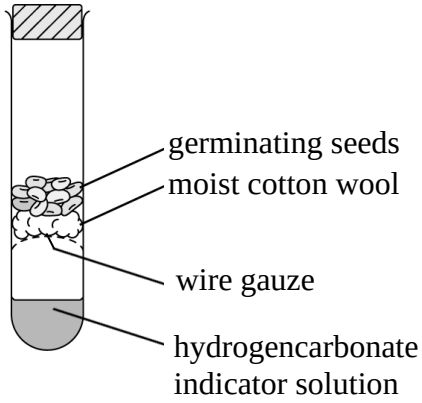
##A##

Skill assessed	Mark
Practical skills	1

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|!|BCEMB0220000031|!| ★ SI

In an investigation, some germinating seeds were put in a boiling tube as shown in the following diagram.



The hydrogencarbonate indicator solution was red at the beginning of the experiment. Which of the following shows the colour of the indicator solution after three hours and the ***most*** suitable interpretation?

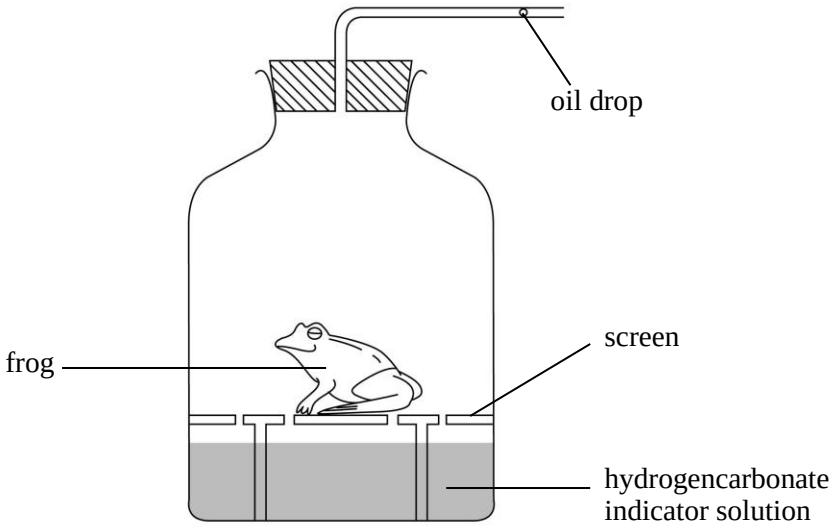
	Colour after three hours	***Interpretation***
A.	red	The rate of photosynthesis equals the rate of respiration.
B.	purple	The rate of photosynthesis is higher than the rate of respiration.
C.	yellow	The seeds carry out respiration and give out carbon dioxide.
D.	yellow	The rate of respiration is higher than the rate of photosynthesis.

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0220000032|!| ★ **SI**

Directions: The following two questions refer to the set-up below, which was used to investigate the respiration of a frog.



The hydrogencarbonate indicator solution was red at the beginning of the experiment. Which of the following was ***most*** likely its colour after four hours?

A. red
B. yellow
C. blue
D. purple

##B##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0220000033|!| ★ **SI**

If the hydrogencarbonate indicator solution was replaced by potassium hydroxide solution, the oil drop would have moved inwards at the end of the experiment. Which of the following was/were the possible cause(s) for this?

(1) Oxygen was taken up by the frog.
(2) Carbon dioxide was absorbed by the potassium hydroxide solution.
(3) Oxygen was absorbed by the potassium hydroxide solution.

A. (1) only
B. (2) only
C. (1) and (2) only
D. (1) and (3) only

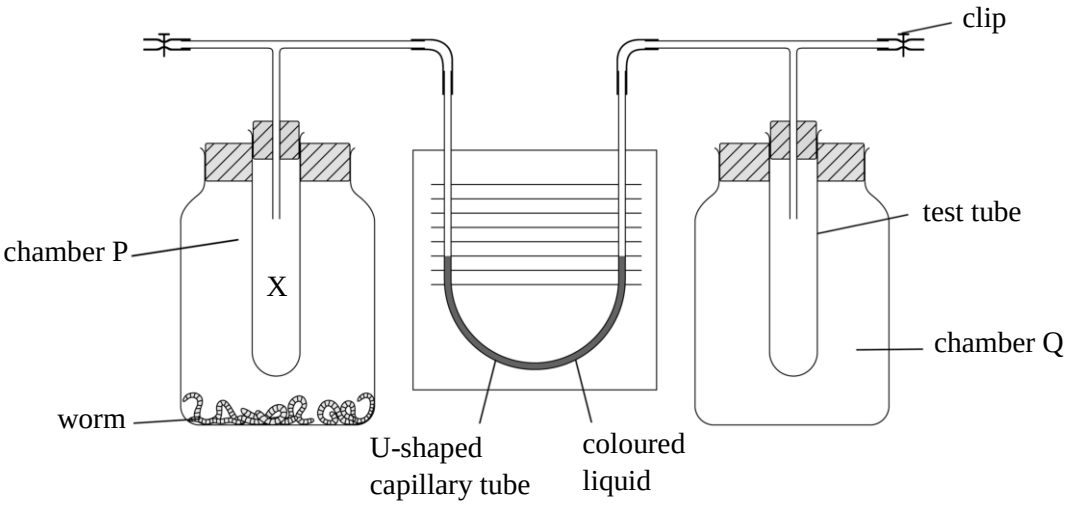
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##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0220000034|!| ★ SI

Directions: The following three questions refer to the diagram below, which shows a differential air thermometer that was used to investigate the respiration of living organisms.



The above apparatus was not set up properly. Which of the following correctly states and explains a correction to the set-up?

	Correction	***Explanation***
A.	Replacing the worms with germinating seeds	Germinating seeds can respire anaerobically.
B.	Adding soda lime to chamber P	Soda lime can absorb carbon dioxide.
C.	Wrapping the chambers with cotton wool	This can reduce heat loss to the surroundings.
D.	Replacing the worms with larger mammals	More carbon dioxide is produced by larger mammals

##C##

Skill assessed	Mark
Practical skills	1

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|!|BCEMB0220000035|!| ★ SI

The differential air thermometer was used to demonstrate that during respiration the worms

A. take in oxygen.
B. release carbon dioxide.
C. release heat.
D. use up food substances.

##C##

Skill assessed	Mark
Practical skills	1

|!|BCEMB0220000036|!| ★ SI

One hour after the correction, the coloured liquid rose in the right arm of the U-shaped capillary tube. Which of the following is ***not*** a relevant explanation for the result?

A. The worms released carbon dioxide in chamber P.
B. The heat released by the worms warmed the air in test tube X.
C. The temperature in chamber P was higher than that in chamber Q.
D. The air in test tube X expanded and pushed the coloured liquid down in the left arm.

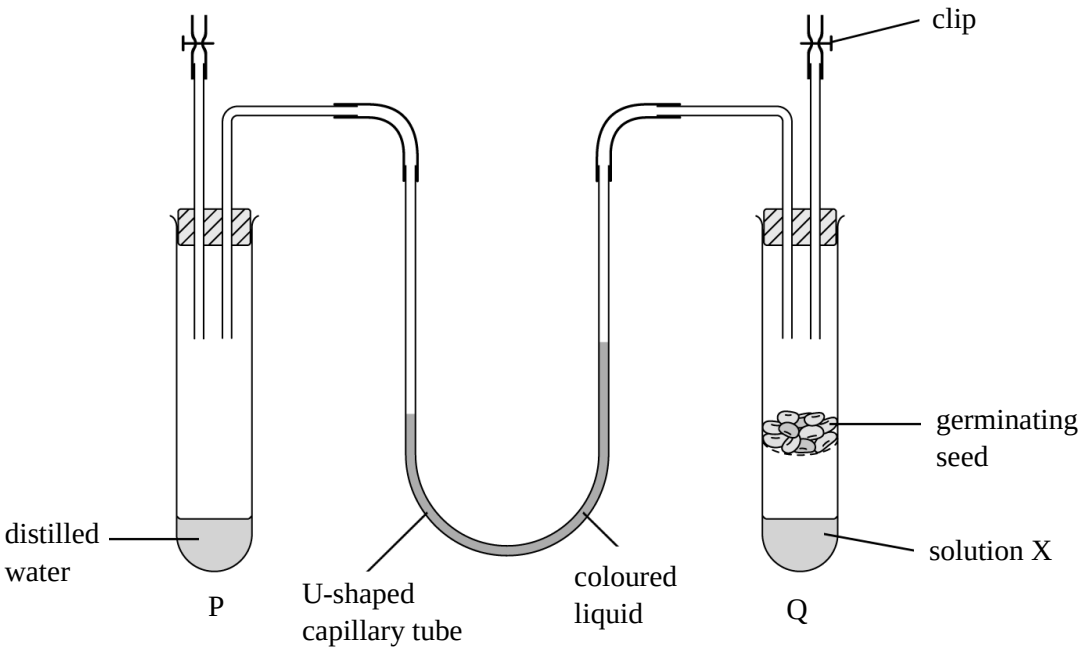
##A##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMB0220000037|!| ★ SI

Directions: The following two questions refer to the diagram below, which shows a set-up used to investigate the respiration of germinating seeds.



After two hours, the liquid level in the U-shaped capillary tube rose in the right arm. What is solution X?

A. hydrogencarbonate indicator solution
B. lime water
C. potassium hydroxide solution
D. distilled water

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0220000038|!| ★ SI

Which of the following account for the rise in the liquid level in the right arm?

(1) absorption of oxygen by the germinating seeds
(2) absorption of carbon dioxide by solution X
(3) release of heat by the germinating seeds

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

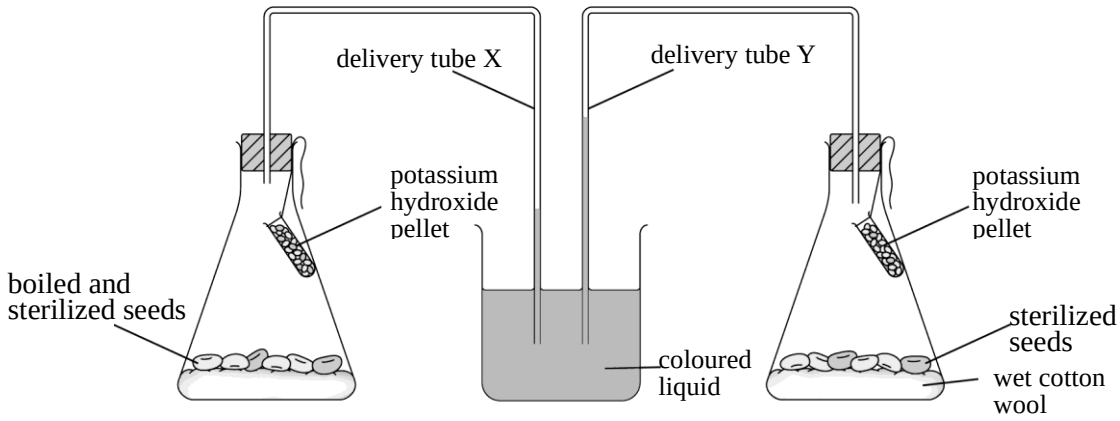
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##A##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0220000039|!| ★ SI

The diagram below shows an experimental set-up used to measure the uptake of oxygen by germinating seeds.



The height of the coloured liquid in tubes X and Y were measured at the beginning of the experiment and 20 minutes later, as recorded in the table below.

Height of coloured liquid	Initial	Final
Delivery tube X	5 cm	5 cm
Delivery tube Y	5 cm	9 cm

Given that the cross-sectional area of the delivery tubes is 0.25 cm^2. The volume of oxygen taken up by the germinating seeds during the experiment was

A. 1 cm^3.
B. 1.25 cm^3.
C. 2.25 cm^3.
D. 3.5 cm^3.

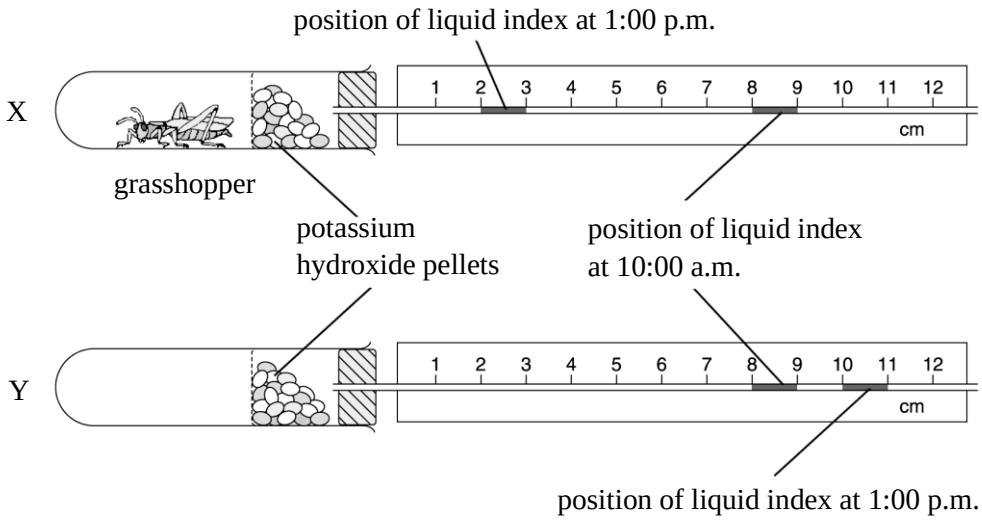
□

##A##

Skill assessed	Mark
Mathematical skills	1

|!|BCEMA0220000040|!| ★★ **SI**

Directions: The following two questions refer to the diagram below, which shows two set-ups used to investigate the respiration of a grasshopper.



Which of the following statements correctly explain(s) the movement of the liquid index in set-up Y?

(1) There was a change in the atmospheric pressure.
(2) There was a change in the surrounding temperature.
(3) The potassium hydroxide pellets gave out oxygen.

A. (1) only
B. (2) only
C. (1) and (2) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMA0220000041|!| ★★ **SI**

The internal cross-sectional area of the glass tubes was 0.3 cm^2. The rate of oxygen consumption of the grasshopper was

A. 0.4 cm^3 / hr.
B. 0.6 cm^3 / hr.
C. 0.8 cm^3 / hr.
D. 1.2 cm^3 / hr.

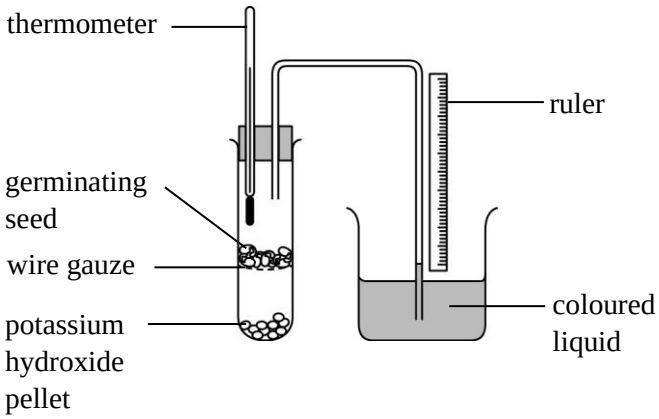
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##C##

Skill assessed	Mark
Mathematical skills	1

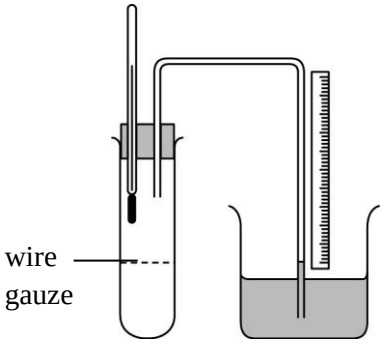
|!|BCEMB0220000042|!| ★ SI

The following diagram shows a set-up used to investigate respiration in germinating seeds.

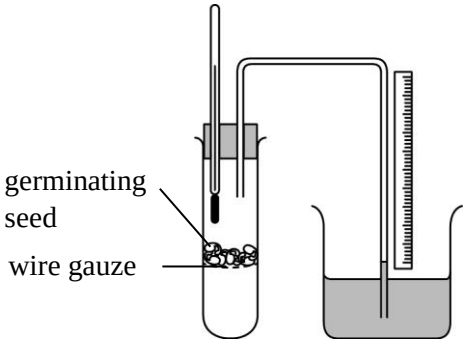


Which of the following shows a suitable control set-up for this investigation?

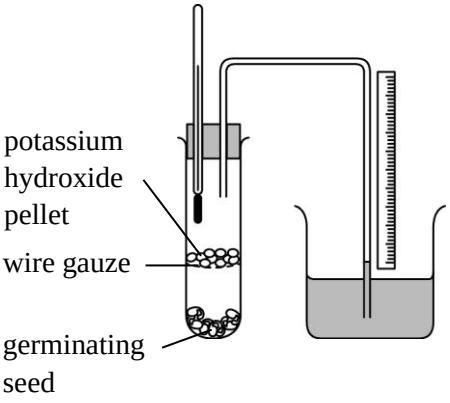
A.



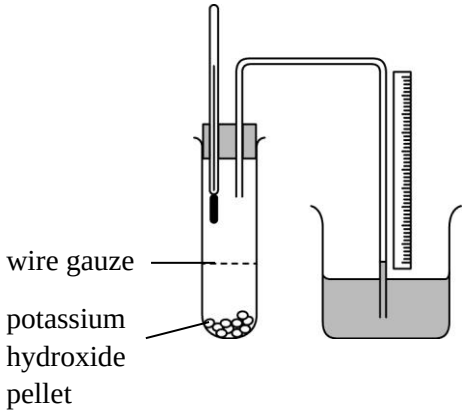
B.



C.



D.



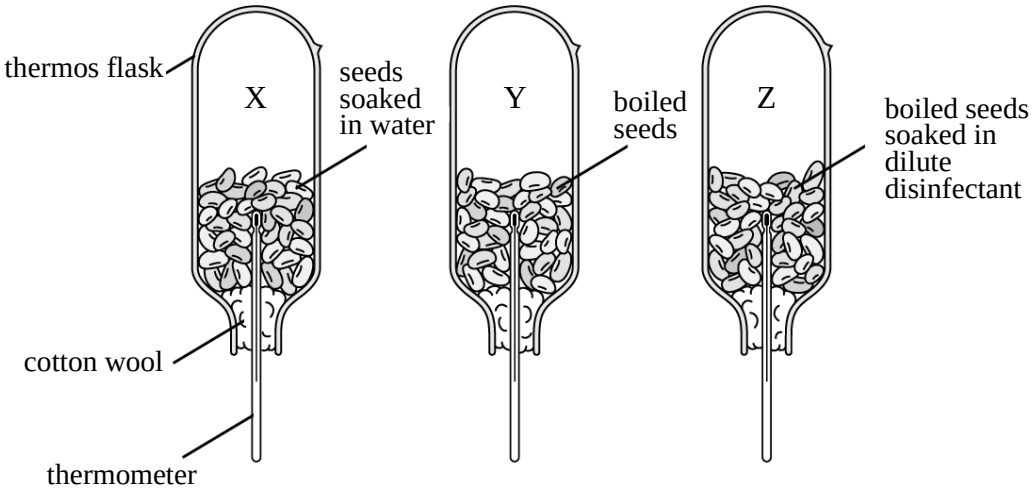
□

##D##

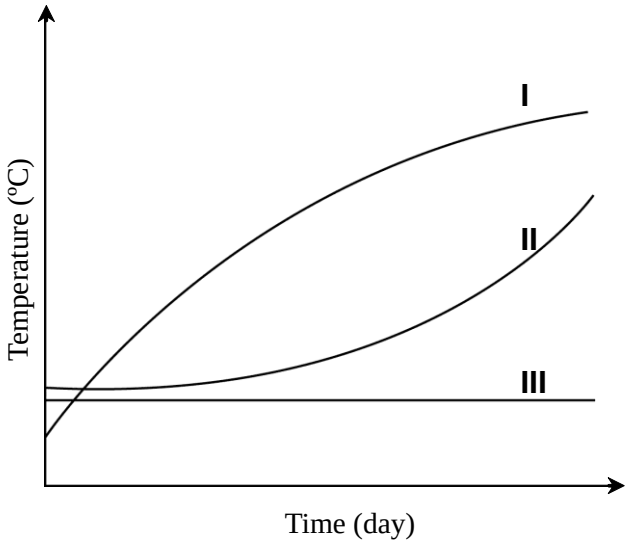
Skill assessed	Mark
Practical skills	1

|!|BCEMB0220000043|!| ★ **SI**

Directions: The following two questions refer to the diagram below, which shows three set-ups used to investigate respiration.



The changes in temperature with time in thermos flasks X, Y and Z are plotted in the graph below.



Which of the following combinations correctly shows the curves that represent the temperature changes in thermos flasks X, Y and Z respectively?

	X	***Y***	***Z***
A.	I	II	III
B.	I	III	II
C.	II	I	III
D.	III	I	II

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##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0220000044|!| ★ SI

The function of the dilute disinfectant used in flask Z was to

A. kill the seeds.
B. soften the seed coat.
C. kill the microorganisms on the surface of the seeds.
D. stop germination of the seeds.

##C##

Skill assessed	Mark
Practical skills	1

|!|BCEMC0220000045|!|

Which of the following are the products of anaerobic respiration in skeletal muscle cells?

(1) ATP
(2) carbon dioxide
(3) lactic acid

A. (1) and (2) only
B. (2) and (3) only
C. (1) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0220000046|!| ★

Which of the following take(s) place in a man's body when he is running a 100 m race?

(1) Aerobic respiration in muscle cells stops.
(2) The blood lactic acid concentration increases.
(3) The blood oxygen level decreases.

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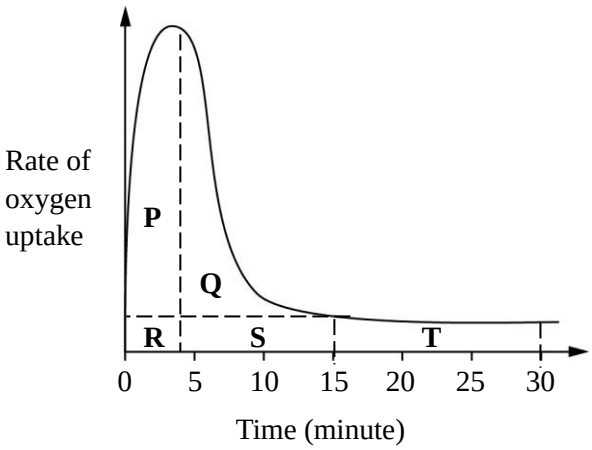
A. (3) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0220000047|!| ★

The following graph shows the changes in the rate of oxygen uptake of an athlete during and after exercise.



Which of the following represents the oxygen debt developed during exercise?

A. area Q
B. area P + area Q
C. area Q + area S
D. area R + area S + area T

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0220000048|!|

Which of the following statements about anaerobic respiration in yeasts are correct?

(1) It involves the Krebs cycle.
(2) It produces ethanol.
(3) Its ATP yield comes solely from glycolysis.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0220000049|!| ★

In alcoholic fermentation, chemical energy in a glucose molecule is converted to

(1) heat energy.
(2) chemical energy in ATP.
(3) chemical energy in alcohol.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

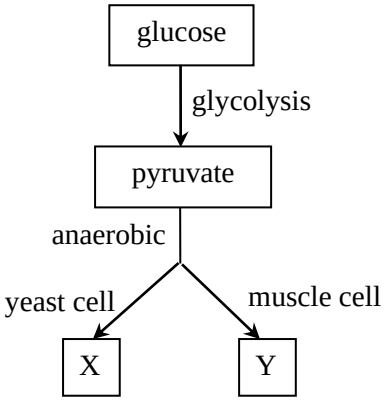
##D##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0220000050|!|

The diagram below shows the anaerobic pathways in yeast and muscle cells.



Which of the following combinations correctly identifies X and Y?

	X	***Y***
A.	ethanol	lactic acid and CO_2
B.	ethanol and CO_2	lactic acid
C.	lactic acid	ethanol
D.	lactic acid and CO_2	ethanol and CO_2

□

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0220000051|!| ★

Which of the following substances is produced as a final product in both alcoholic fermentation and lactic acid fermentation?

A. carbon dioxide
B. water
C. pyruvate
D. ATP

□

##D##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMB0220000052|!| ★

We continue to breathe deeply and rapidly for a while after exercise. What is the significance of this?

A. To supply more oxygen to muscle cells for a faster rate of aerobic respiration
B. To supply more oxygen for liver cells to break down lactic acid
C. To stop anaerobic respiration in mucle cells
D. To prevent the formation of alcohol from the anaerobic respiration in mucle cells

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0220000053|!| ★

Which of the following comparisons between lactic acid fermentation and alcoholic fermentation is correct?

	Lactic acid fermentation	***Alcoholic fermentation***
A.	More energy is released.	Less energy is released.
B.	Carbon dioxide is produced.	No carbon dioxide is produced.
C.	NADH is formed.	No NADH is formed.
D.	It occurs in skeletal muscles.	It occurs in germinating seeds.

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0220000054|!| ★

Which of the following comparisons between aerobic respiration and anaerobic respiration in yeast cells is ***incorrect***?

	Aerobic respiration	***Anaerobic respiration***
A.	It occurs in the cytoplasm and mitochondria.	It occurs in mitochondria only.
B.	Glucose is completely broken down.	Glucose is not completely broken down.
C.	End products are CO_2 and H_2O.	End products are CO_2 and ethanol.
D.	38 molecules of ATP are produced using one molecule of glucose.	2 molecules of ATP are produced using one molecule of glucose.

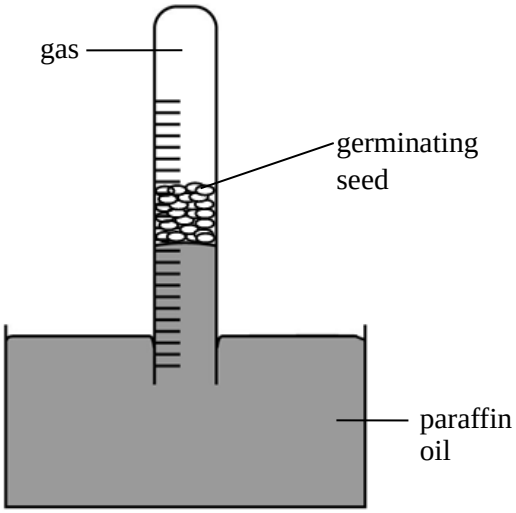
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##A##

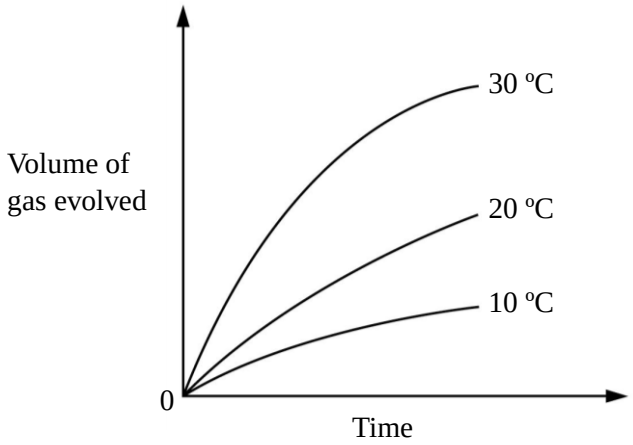
Skill assessed	Mark
Applying concepts	1

|!|BCEMB0220000055|!| ★ SI

Directions: The following three questions refer to the following experimental set-up for investigating the respiration of germinating seeds at different temperatures by comparing the volume of gas collected.



The graph below shows the results.



Which of the following is the main reason for using paraffin oil in this experiment?

A. It supplies nutrients to the seeds.
B. It can provide an anaerobic condition.
C. The gas released by the seeds can be collected.
D. It protects the seeds against bacterial infection.

□

##B##

Skill assessed	Mark
Practical skills	1

|!|BCEMB0220000056|!| ★ SI

Which of the following combinations correctly show a test for the gas collected and the identity of the gas?

	Test for the gas	*Identity of the gas*
A.	a glowing splint	oxygen
B.	lime water	carbon dioxide
C.	hydrogencarbonate indicator solution	oxygen
D.	anhydrous cobalt chloride paper	water vapour

##B##

Skill assessed	Mark
Practical skills	1

|!|BCEMB0220000057|!| ★ SI

Which of the following correctly explain(s) the results shown in the graph?

(1) A higher temperature speeds up the respiration rate of germinating seeds.
(2) A higher temperature causes an expansion of the gas.
(3) A lower temperature leads to reduced activity of the enzymes involved in respiration.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

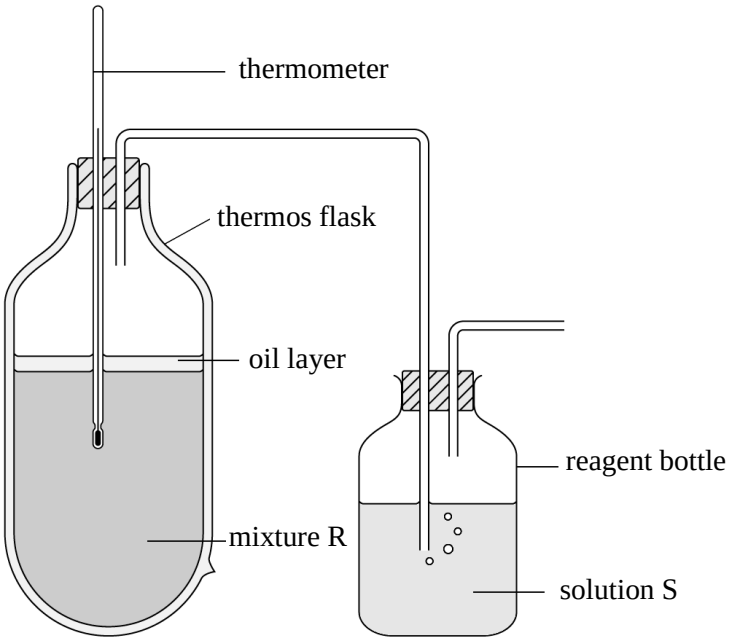
##D##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0220000058|!| ★ **SI**

Directions: The following three questions refer to the diagram below which shows an experimental set-up used to demonstrate alcoholic fermentation.



The function of the oil layer was to stop

A. carbon dioxide from leaving mixture R.
B. carbon dioxide from entering mixture R.
C. oxygen from leaving mixture R.
D. oxygen from entering mixture R.

##D##

Skill assessed	Mark
Practical skills	1

|!|BCEMB0220000059|!| ★ **SI**

In order to show that carbon dioxide was released from mixture R, solution S should be

A. sodium hydrogencarbonate solution.
B. lime water.
C. potassium hydroxide solution.
D. calcium chloride solution.

##B##

Skill assessed	Mark
Practical skills	1

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|!|BCEMB0220000060|!| ★ SI

The process of fermentation would eventually stop because

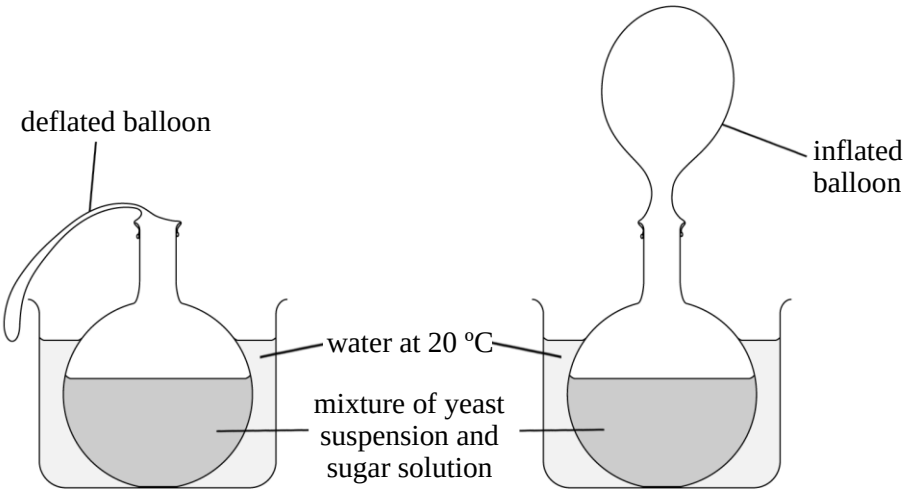
A. the yeast cells died due to a lack of oxygen.
B. the yeast cells died due to high temperature.
C. the yeast cells died due to overcrowding.
D. the yeast cells were killed by the accummulation of the fermentation products.

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMA0220000061|!| ★★ SI

Directions: The following three questions refer to the diagram below which shows the set-up used to investigate the action of yeast and the experimental results.



The balloon was inflated at the end of the experiment because of the

A. expansion of the air inside the flask and the balloon.
B. evaporation of water from the mixture in the flask.
C. evaporation of ethanol from the mixture in the flask.
D. production of gas from the mixture in the flask.

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMA0220000062|!| ★★ **SI**

The gas in the inflated balloon can

A. turn hydrogencarbonate indicator solution red.
B. turn lime water milky.
C. relight a glowing splint.
D. turn anhydrous cobalt chloride paper blue.

##B##

Skill assessed	Mark
Practical skills	1

|!|BCEMA0220000063|!| ★★ **SI**

Which of the following would ***not*** lead to a bigger size of the balloon at the end of the experiment?

A. increasing the amount of sugar in the mixture
B. increasing the amount of yeast in the mixture
C. increasing the temperature of water bath to 30 °C.
D. increasing the amount of water in the mixture

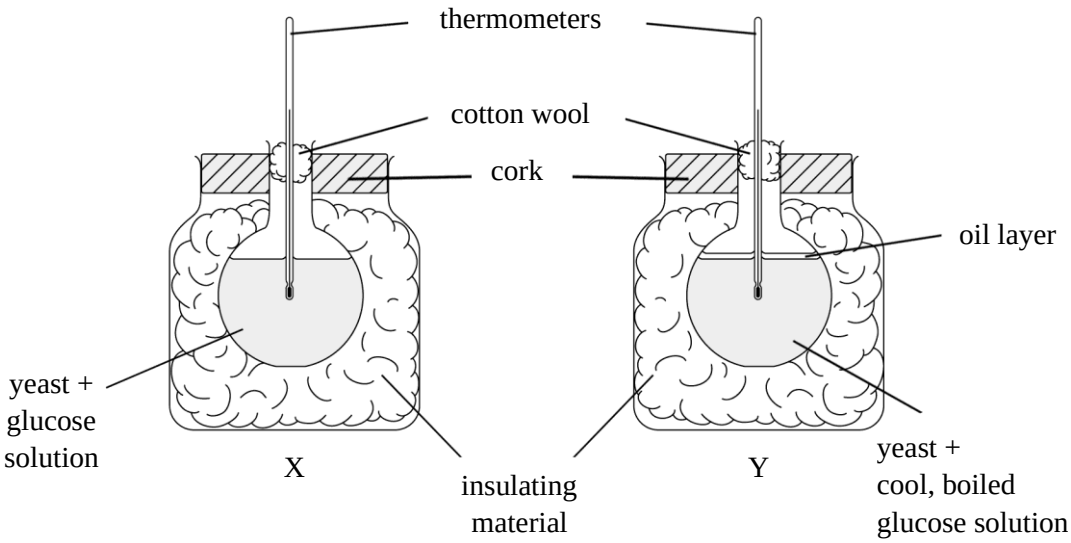
##D##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMB0220000064|!| ★ SI

Directions: The following two questions refer to the diagram below which shows two set-ups used to investigate the respiration of yeast.



What were the products of respiration in set-ups X and Y?

	X	***Y***
A.	carbon dioxide and water	carbon dioxide and ethanol
B.	carbon dioxide and water	lactic acid
C.	carbon dioxide and ethanol	lactic acid
D.	carbon dioxide and ethanol	carbon dioxide and ethanol

##A##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0220000065|!| ★ SI

Initially, the temperatures of both set-ups were 20 °C. Which of the following combinations shows the ***most probable*** temperatures in the set-ups after two hours?

	X	***Y***
A.	21 °C	23 °C
B.	20 °C	23 °C
C.	23 °C	21 °C
D.	20 °C	20 °C

##C##

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Skill assessed	Mark
Applying concepts	1

|!|BCEMC0220000066|!|

What is the role of yeast in bread-making?

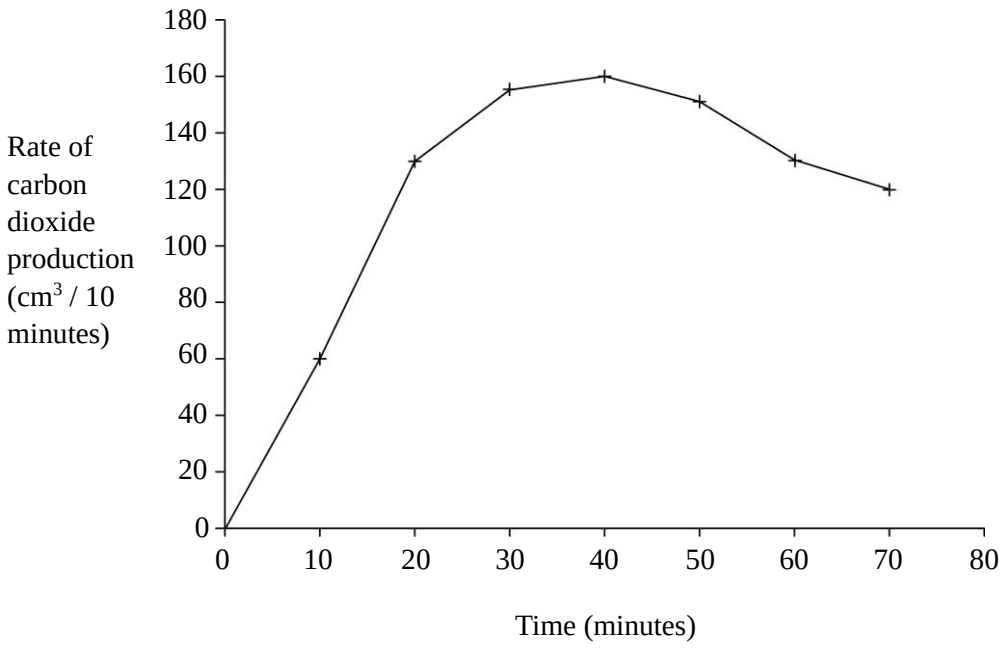
A. Yeast can break down proteins in the dough and make the bread soft.
B. Yeast can convert sugars in the dough to ethanol and carbon dioxide which causes the dough to rise.
C. Yeast carries out alcoholic fermentation and produces heat which causes the dough to rise.
D. Yeast carries out alcoholic fermentation and produces ethanol which makes the bread soft.

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0220000067|!| **SI**

Peter carried out an investigation to measure the rate of carbon dioxide production in a bread dough at 35 °C. He prepared the dough by mixing flour, sugar, water and yeast. The following graph shows the results of the investigation.



Which of the following causes the decrease in the rate of carbon dioxide production in the bread dough after 40 minutes?

(1) the decrease in the sugar content of the dough
(2) the increase in the yeast population
(3) the build up of ethanol which is toxic to yeast

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

□

##B##

Skill assessed	Mark
Applying concepts	1